

A Lipofuscin-like Pigment in the Kidneys of Estrogen-Treated Rats

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A golden-brown pigment is deposited in the cells of the proximal convoluted tubule of the rat kidney after long-term treatment with high doses of estrogen. Dissection of the tubule reveals that the pigment is deposited at random in this segment and is never observed in other portions of the nephron. Histochemical tests suggest that the pigment may be a variant of the lipofuscins. It is not anatomically associated with sites of acid phosphatase activity. Iron in the ferric form increases in amount as pigment is deposited, but is not uniformly noted at pigment loci.

WHENEVER 17- β estradiol in the form of 25 mg pellets is implanted subcutaneously in either male or female rats, a golden pigment forms within the cells of the proximal convoluted tubule of the kidney. The pigment has been observed in all rats treated in this manner with estrogen during the course of experiments in mammary tumorigenesis that have been performed during the past 15 years. The rats may be housed singly or in groups, and are allowed free access to food and water. Although in most of the experiments the estrogen was implanted between the ages of 30 to 60 days, implantation in rats 4 months of age also resulted in pigment formation. The pigment first becomes visible eight weeks after the inception of treatment, and by 18 weeks, is sufficiently concentrated to cause an olive-brown discoloration of the renal cortex. Microscopic particles of translucent golden-brown material may reach 6 μ to 8 μ in diameter (Fig 1). Dissection of the tubules

after maceration with concentrated hydrochloric acid confirms the impression, derived from microscopic studies of histologic sections, that the pigment is restricted to the proximal convoluted tubule. It appears to be randomly deposited with no gradient apparent as to particle size or concentration (Fig 2). The reactions of the pigment to stains and chemicals are listed in the Table.

Because prolonged administration of estrogen inhibits body growth, the possibility was considered that the pigment might have been deposited as a result of vitamin E deficiency. To test this hypothesis, two groups of ten rats each were established. All rats were housed singly in wire mesh cages and fed pellets (Rockland) ad libitum. To both groups, at age 50 days, 25 mg pellets of estradiol were implanted subcutaneously, and one group received, in addition, 1 cc of wheat germ oil daily by stomach tube, for the ten weeks that the estrogen pellets remained in situ. Pigment was found in the kidneys of both groups of rats, indicating that its formation was not due to vitamin E deficiency.

Iron in the ferric form is associated with the pigment, and increases in amount as the quantity of pigment increases, but it can be dissolved out after incubation for 18 hours in concentrated hydrochloric acid without substantially altering the appearance of the pigment.

Since chromophobe tumors of the pituitary invariably accompany prolonged estrogen administration, the possibility existed that the pigment might be a melanin, produced in large quantities due to excessive stimulation by melanocyte stimulating hormone (MSH). However, in tissues examined for a pharmaceutical company, a similar pigment was

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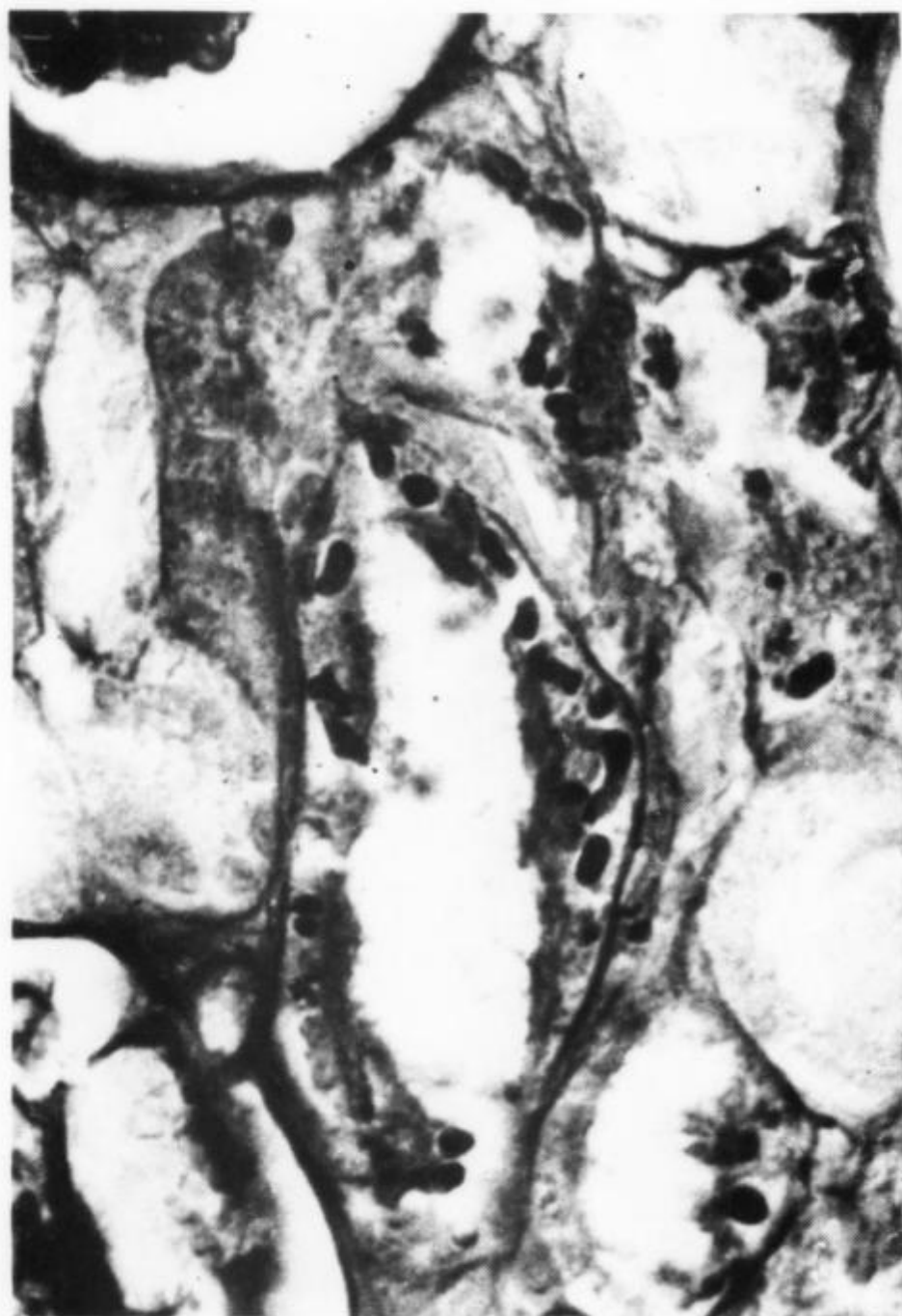


Fig 1.—Pigment particles in the cells of the proximal convoluted tubules of the rat kidney (PAS, $\times 750$ reduced about 40%).

found in the proximal convoluted tubules of the kidneys of rats given a drug that had an estrogenic effect on the uterus and ovaries, but which did not affect the pituitary. In addition, staining reactions for melanin, such as Schmorl's reaction, and Lillie's ferrous sulfate test, are negative.

Comment

Two references to a similar pigment have appeared recently: the "cloisonné" kidneys

of castrated male white angora goats are characterized by a lipofuscin-like pigment which is deposited in the basement membranes of the proximal convoluted tubules¹; and black kidney disease of sheep, found in Australia, which differs from the above, in that the pigment is located within the cells of the proximal convoluted tubule.² The induction by estrogen of pigment in a similar location in the kidneys of rats suggests that the "spontaneous" cases in animals might be due to the prolonged ingestion of some dietary substance. A similar, but not identical pigment (lipofuscin) occurs in the brain, heart, and liver of aged individuals.³ Many pigmentations may be the result of a discrepancy between the size of an intrinsic chemical load at a particular intracellular locus, and the amount of intrinsic chemical or enzymatic potential available for disposing of that load. Perhaps in the cells of the aged, the intracellular mechanism has deteriorated, or is not replaced with sufficient speed to cope with the normal daily needs, whereas in the estrogen kidney the overload of metabolite overwhelms the clearing mechanism, resulting in the deposition of pigment.

Although the lipofuscins in brain and liver appear to be anatomically associated with the cytosome,^{4,5} (an organelle thought by some to perform an excretory function in the cell)⁶ acid phosphatase stains of estrogen treated kidneys fail to demonstrate acid phosphatase activity at the site of pigment formation (the Table).

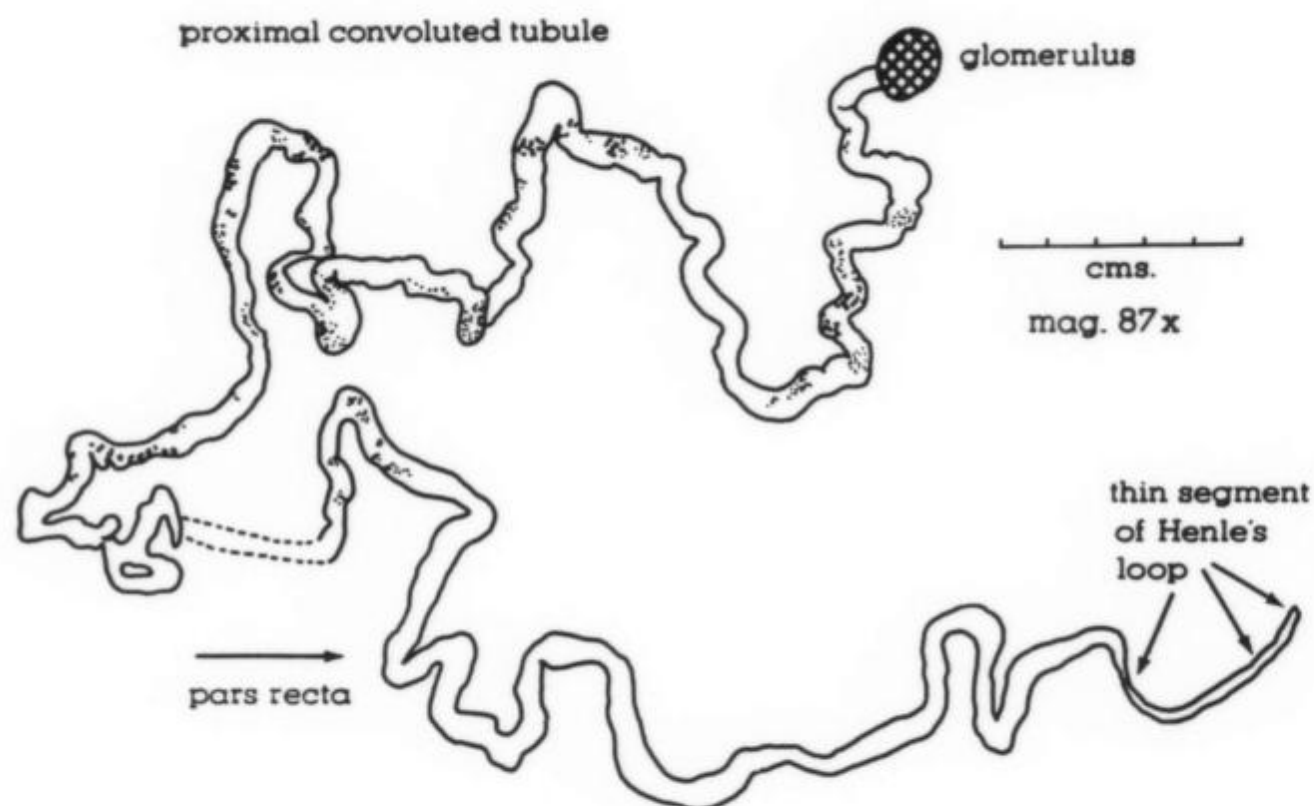


Fig 2.—Camera lucida tracing of nephron from rat treated with estrogen showing distribution of pigment in proximal convoluted tubule ($\times 43.5$).

*Reactions to Chemicals and Stains of Intracellular Pigment
in the Proximal Convoluted Tubules of Estrogen Treated Rats*

1. Organic solvents (alcohol, xylene, Folch reagent, ether)	Insoluble
2. Aqueous solvents (water, hydrochloric, and sulfuric acid; sodium hydroxide)	Insoluble
3. Oxidizing agents Nitric acid (concentrated), 24 hr Hydrogen peroxide (30%), 24 hr Potassium permanganate (0.1%), 24 hr	Color fades
4. Stains Sudan IV—frozen and paraffin sections Osmic acid 1% 24 hr—paraffin sections Hematoxylin Eosin Ziehl-Nielsen for acid fastness PAS Schmorl's reaction ⁷ Lillie's ferrous sulfate test for melanin ⁷ , (p 919) Nile blue sulfate Masson-Fontana method for melanin ⁷ , (p 919) (6 hr 24 hr) Acid phosphatase (Wachstein after Gomori) ⁸ & azo dye method with α -naphthol AS-MX phosphate ⁹ Perls' reaction for ferric ion Turnbull blue for ferrous ion	Negative Negative Negative Negative Variable Weak Negative Negative Positive Negative Positive There is no particular association between pigment & zones of acid phosphatase activity A definite association exists between quantity of stainable iron & the amount of pigment present but iron is not necessarily deposited at pigment sites Negative No fluorescence
5. Ultraviolet	

The precise nature of the pigment is still, as it is for most pigments, a mystery, since staining and chemical reactions fail to identify it with any single group, although its basophilia under Nile blue sulfate and the fact that it bleaches when exposed to oxidizing agents, suggest that it should be classified

in the category of pseudomelanins and lipofuscins.

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